

Preclass Learning: Databases and SQL (ICA 16)

1. Relational Database Concepts

- **Database Management System (DBMS):** Software that manages the database (e.g., SQLite, PostgreSQL).
- **Table:** The primary storage unit, organized into rows (records) and columns (fields/attributes).
- **Schema:** The logical structure or blueprint of the entire database, including table names, column names, and their data types.
- **Indexing:** Special lookup tables used by the DBMS to speed up data retrieval operations.

2. Core SQL Queries (Data Retrieval)

SQL is a declarative language, meaning you describe *what* data you want, not *how* to get it.

- **Basic Structure:**
SELECT column_list
FROM table_name
WHERE condition
GROUP BY column
ORDER BY column
LIMIT N;
- **SELECT:** Specifies the columns to return.
- **WHERE:** Filters rows based on a condition (e.g., WHERE age > 30).
- **GROUP BY:** Aggregates data into groups (e.g., calculate the average score *per team*).
- **JOIN:** Combines data from two tables based on a shared key (see ICA 7).

3. R/Python to Database Connectivity

- **Drivers:** Software components (like ODBC/JDBC) that allow R or Python to communicate with a specific DBMS.
- **dbplyr (R):** A package that translates R's dplyr commands into efficient SQL, allowing the database to perform the heavy lifting before data is transferred to R's memory.